

PROJECT

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CLOUD SECURITY WITH AWS IAM

SECURITY SERVICE

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Objective

This project is designed to help you understand how to secure AWS resources using Identity and Access Management (IAM). You'll set up a virtual server using EC2 and control access to it by defining permissions.

You will also learn to create an account alias for better login experience, set up user groups for interns, and test restricted access, ensuring that users only perform the actions they're allowed to.

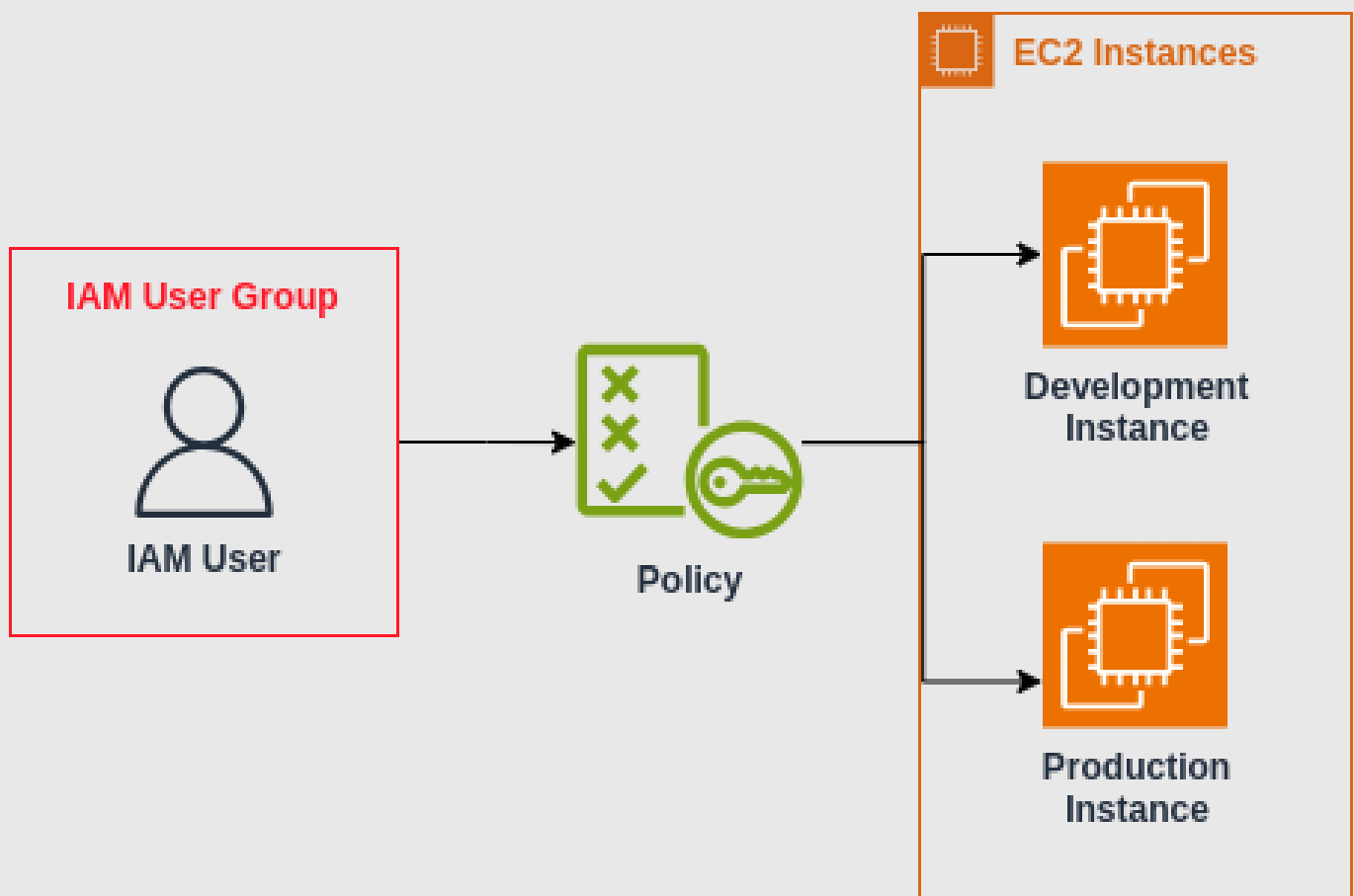
Services

Amazon EC2 - Launches and manages virtual servers to simulate real cloud computing environments.

AWS IAM - Creates users, groups, and permission policies to control access to AWS resources.

Account Alias - Customizes the IAM login URL to make it user-friendly and easy to remember.

Architecture Diagram



An EC2 instance is launched to serve as the main resource. A custom IAM policy is created and attached to a group assigned to interns. IAM users are created and placed in this group. The account alias customizes the login link, and once users log in, their access is tested, verifying they can only interact with EC2 as permitted by the policy, ensuring cloud access remains secure and role based.

Steps to Implement

1. Launch an EC2 Instance:

Open the EC2 console and click “Launch Instance.” Choose a suitable AMI (like Amazon Linux), select an instance type (e.g., t2.micro), configure network settings, and launch the instance using a new or existing key pair. This instance will be used to test access control later.

2. Create a Custom IAM Policy:

Go to the IAM console and select “Policies,” then click “Create policy.” Use the visual editor to allow specific EC2 actions like StartInstances, StopInstances, and DescribeInstances. Name the policy (e.g., EC2AccessPolicy) and save it for attaching to a user group.

3. Set Up Account Alias, User Group, and Users:

In the IAM dashboard, set a custom account alias for easy login (e.g., company intern). Create a new user group (e.g., InternGroup) and attach the custom policy. Then add IAM users (e.g., intern1) and assign them to the group, enabling console access with passwords.

4. Test Access with IAM User:

Log in using the IAM user credentials and custom alias URL. Try accessing the EC2 service and verify that the user can only perform allowed actions like viewing, starting, or stopping the instance ensuring the policy restrictions work as intended.

Story Time

Amazon EC2 (The Cloud Beast Arena):

Imagine Amazon EC2 as your powerful beast arena where digital warriors (servers) come to life. You summon a beast, give it strength (CPU), memory (RAM), and a place to live (instance), and it's ready to carry out your tasks in the vast cloud kingdom.

AWS IAM (The Access Gatekeeper):

Think of AWS IAM as the wise gatekeeper of your cloud castle. It controls who gets to enter, what they can do, and which rooms they can explore. With keys (users), roles (armor), and scrolls (policies), IAM makes sure only the right heroes touch the right tools.

Account Alias (The Secret Door Name):

The Account Alias is your cloud castle's custom front gate. Instead of a boring random link, you get to name your magical entry point, like "intern portal", so your adventurers know exactly where to go when starting their AWS journey!

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THANK YOU!

